LAB-ASSIGMENT:3
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Objectives

- **To understand the use of COMMIT and ROLLBACK commands in transaction control.**
- **To use ORDER BY and GROUP BY clauses in SQL queries.**
- **To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.**
- **To understand and implement JOIN operations in SQL.**

DDL Questions (Data Definition Language)

1. **Create the TEACHER table with the following attributes: (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).**

TEACHER TABLE

```
CREATE TABLE TEACHER (
  Name VARCHAR(50),
  DeptNo INT,
  Date_of_Joining DATE,
  DeptName VARCHAR(50),
  Location VARCHAR(50),
  Salary INT
);
```

2. **Create the SALES table with the following attributes: (Sales_No, Salesname, Branch, Salesamount, DOB).**

SALES TABLE

```
CREATE TABLE SALES (
  Sales_No INT PRIMARY KEY,
  Salesname VARCHAR(50),
  Branch VARCHAR(50),
  Salesamount INT,
  DOB DATE
);
```

3. **Create the DEPARTMENTS table with the following attributes:**
   (dept_no, dept_name, dept_location).

DEPARTMENTS TABLE

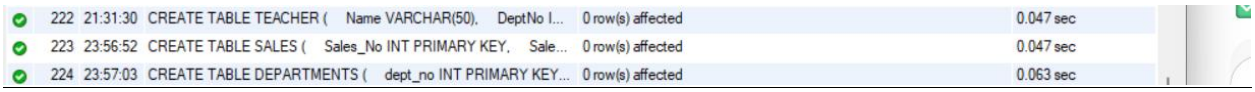
```
CREATE TABLE DEPARTMENTS (
  dept_no INT PRIMARY KEY,
  dept_name VARCHAR(50),
  dept_location VARCHAR(50)
);
```

4.Create the EMPLOYEES table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

EMPLOYEES Table

```
CREATE TABLE EMPLOYEES (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  emp_salary INT,
  dept_no INT,
  FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

SCREENSHOTS:



222	21:31:30	CREATE TABLE TEACHER (Name VARCHAR(50), DeptNo I...	0 row(s) affected	0.047 sec
223	23:56:52	CREATE TABLE SALES (Sales_No INT PRIMARY KEY, Sale...	0 row(s) affected	0.047 sec
224	23:57:03	CREATE TABLE DEPARTMENTS (dept_no INT PRIMARY KEY...	0 row(s) affected	0.063 sec

DML Questions (Data Manipulation Language)

From TEACHER Table

1. **Insert five records into the TEACHER table.**

```sql
INSERT INTO TEACHER VALUES
('Anita', 1, '2015-06-10', 'Mathematics', 'Delhi', 40000),
('Rahul', 2, '2018-07-15', 'Commerce', 'Mumbai', 35000),
('Priya', 1, '2016-08-20', 'Mathematics', 'Delhi', 42000),
('Karan', 3, '2017-09-25', 'Science', 'Chennai', 38000),
('Sneha', 2, '2019-10-30', 'Commerce', 'Mumbai', 36000);
SELECT*FROM TEACHER;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita	1	2015-06-10	Mathematics	Delhi	40000
Rahul	2	2018-07-15	Commerce	Mumbai	35000
Priya	1	2016-08-20	Mathematics	Delhi	42000
Karan	3	2017-09-25	Science	Chennai	38000
Sneha	2	2019-10-30	Commerce	Mumbai	36000

2. **Give an increment of 25% salary to teachers working in the Mathematics department.**

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
select*FROM TEACHER;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita	1	2015-06-10	Mathematics	Delhi	50000
Rahul	2	2018-07-15	Commerce	Mumbai	35000
Priya	1	2016-08-20	Mathematics	Delhi	52500
Karan	3	2017-09-25	Science	Chennai	38000
Sneha	2	2019-10-30	Commerce	Mumbai	36000

3. **Give an increment of 15% salary to teachers working in the Commerce department.**

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
SELECT*FROM TEACHER;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita	1	2015-06-10	Mathematics	Delhi	50000
Rahul	2	2018-07-15	Commerce	Mumbai	40250
Priya	1	2016-08-20	Mathematics	Delhi	52500
Karan	3	2017-09-25	Science	Chennai	38000
Sneha	2	2019-10-30	Commerce	Mumbai	41400

From SALES Table

1. **Insert five records into the SALES table.**

```sql
INSERT INTO SALES VALUES
(101, 'Amit', 'Delhi', 50000, '1999-12-21'),
(102, 'Rohit', 'Mumbai', 60000, '2000-05-15'),
(103, 'Sohan', 'Delhi', 45000, '1998-12-10'),
(104, 'Kunal', 'Chennai', 70000, '1997-03-20'),
(105, 'Raj', 'Mumbai', 55000, '1996-12-05');
SELECT*FROM SALES;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wr

Sales_No	Salesname	Branch	Salesamount	DOB
101	Amit	Delhi	50000	1999-12-21
102	Rohit	Mumbai	60000	2000-05-15
103	Sohan	Delhi	45000	1998-12-10
104	Kunal	Chennai	70000	1997-03-20
105	Raj	Mumbai	55000	1996-12-05
NULL	NULL	NULL	NULL	NULL

2. **Calculate the total sales amount in each branch.**

```sql
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	Total_Sales
Delhi	95000
Mumbai	115000
Chennai	70000

3. **Calculate the average sales amount in each branch.**

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	Average_Sales
Delhi	47500.0000
Mumbai	57500.0000
Chennai	70000.0000

4. **Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.**

```
SELECT Salesname,
DATE_FORMAT(DOB,'%d-%b-%y') AS Formatted_DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Salesname	Formatted_DOB
Amit	21-Dec-99
Sohan	10-Dec-98
Raj	05-Dec-96

5. **Display the Salesname and DOB of salesmen in alphabetical order of the month.**

```
SELECT Salesname,
DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Salesname	DOB
Amit	21-Dec-99
Sohan	10-Dec-98
Raj	05-Dec-96
Kunal	20-Mar-97
Rohit	15-May-00

<u>From JOIN Operations</u>

1. **Display employee details and department details where the department numbers are the same in both tables.**

```
SELECT *
FROM EMPLOYEES e
LEFT JOIN DEPARTMENTS d
ON e.DEPARTMENT_ID = d.DEPT_NO;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id	dept_no	dept_name	dept_location
101	Anaya	Singh	anaya@gmail.com	9876543210	2024-01-10	DEV	50000	0.10	201	60	NULL	NULL	NULL
102	Riya	Sharma	riya@gmail.com	9876543211	2023-05-15	HR	40000	0.05	202	70	NULL	NULL	NULL
103	Karan	Mehta	karan@gmail.com	9876543212	2022-03-20	DEV	60000	0.12	201	80	NULL	NULL	NULL

2. **Display the employee name and department name using a LEFT OUTER JOIN.**

```
SELECT e.First_Name, d.dept_name
FROM EMPLOYEES e
LEFT JOIN DEPARTMENTS d
ON e.Department_Id = d.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	dept_name
Anaya	NULL
Riya	Finance
Karan	IT

3. **Display the employee name and department name using a RIGHT OUTER JOIN.**

```
SELECT CONCAT(e.First_Name, ' ', e.Last_Name) AS Employee_Name,
       d.dept_name
FROM EMPLOYEES e
RIGHT JOIN DEPARTMENTS d
ON e.Department_Id = d.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Name	dept_name
NULL	HR
NULL	Finance
NULL	IT
Riya Sharma	Finance
Karan Mehta	IT
NULL	HR

4. **Display the details of employees who draw a salary greater than the average salary.**

```
SELECT *FROM EMPLOYEES
WHERE Salary >(SELECT AVG(Salary) FROM EMPLOYEES);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
103	Karan	Mehta	karan@gmail.com	9876543212	2022-03-20	DEV	60000	0.12	201	80
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

TCL Questions (Transaction Control Language)

1. **Perform the ROLLBACK command on the TEACHER table.**

```
START TRANSACTION;
UPDATE TEACHER
SET Salary = Salary + 5000
WHERE DeptName = 'Mathematics';
ROLLBACK;
select*from teacher;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita	1	2015-06-10	Mathematics	Delhi	50000
Rahul	2	2018-07-15	Commerce	Mumbai	40250
Priya	1	2016-08-20	Mathematics	Delhi	52500
Karan	3	2017-09-25	Science	Chennai	38000
Sneha	2	2019-10-30	Commerce	Mumbai	41400

2. **Perform the COMMIT command on the TEACHER table.**

```sql
START TRANSACTION;

UPDATE TEACHER
SET Salary = Salary + 2000
WHERE DeptName = 'Commerce';
COMMIT;
select*from teacher;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita	1	2015-06-10	Mathematics	Delhi	50000
Rahul	2	2018-07-15	Commerce	Mumbai	42250
Priya	1	2016-08-20	Mathematics	Delhi	52500
Karan	3	2017-09-25	Science	Chennai	38000
Sneha	2	2019-10-30	Commerce	Mumbai	43400

3. **Write a query to implement a SAVEPOINT.**

```sql
START TRANSACTION;
UPDATE TEACHER
SET Salary = Salary + 3000
WHERE DeptName = 'Science';
SAVEPOINT A;
UPDATE TEACHER
SET Salary = Salary + 2000
WHERE DeptName = 'Commerce';
ROLLBACK TO A;
COMMIT;
select*from teacher;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita	1	2015-06-10	Mathematics	Delhi	50000
Rahul	2	2018-07-15	Commerce	Mumbai	42250
Priya	1	2016-08-20	Mathematics	Delhi	52500
Karan	3	2017-09-25	Science	Chennai	41000
Sneha	2	2019-10-30	Commerce	Mumbai	43400

DCL Questions (Data Control Language)

1. **Write a query to grant all privileges on the EMPLOYEES table to DEPARTMENTS.**

```sql
GRANT ALL PRIVILEGES
ON EMPLOYEES
TO 'departments'@'localhost';
SHOW GRANTS FOR 'departments'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Grants for departments@localhost
GRANT USAGE ON *.* TO `departments`@`loc...
GRANT ALL PRIVILEGES ON `dbms_anshika`.`e...

2. **Write a query to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.**

```sql
GRANT SELECT, INSERT
ON EMPLOYEES
TO 'departments'@'localhost';
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
294	01:12:18	SELECT CURRENT_USER() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
295	01:13:26	SHOW GRANTS FOR 'departments'@'localhost'	2 row(s) returned	0.000 sec / 0.000 sec
296	01:13:49	REVOKE ALL PRIVILEGES ON your_database_name.EMPLOYEES...	Error Code: 1147. There is no such grant defined for user 'departmen...	0.000 sec
297	01:17:49	GRANT SELECT, INSERT ON EMPLOYEES TO 'departments'@'lo...	0 row(s) affected	0.016 sec

3. **Write a query to revoke all privileges on the EMPLOYEES table from DEPARTMENTS.**

```
SELECT DATABASE();
GRANT ALL PRIVILEGES
ON dbms_anshika.EMPLOYEES
TO 'departments'@'localhost';

REVOKE ALL PRIVILEGES
ON dbms_anshika.EMPLOYEES
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
297	01:17:49	GRANT SELECT, INSERT ON EMPLOYEES TO 'departments'@'lo...	0 row(s) affected	0.016 sec

4. **Write a query to revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.**

```
REVOKE SELECT, INSERT
ON dbms_anshika.EMPLOYEES
FROM 'departments'@'localhost';

SHOW GRANTS FOR 'departments'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Grants for departments@localhost
GRANT USAGE ON *.* TO `departments`@`loc...
GRANT UPDATE, DELETE, CREATE, DROP, REF...